

Frames, Returns, and Port Power

Exact work balances for epicyclic gears and transformer readouts

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Abstract

An angular displacement, a voltage integral, and a transferred work are different observables. We derive their relations for a parallel-axis epicyclic train, physical shaft readouts, and tapped or isolated coupled windings. Signed port integrals locate the effect of a rotating mechanical frame at every rate-dependent port; a common electrical voltage offset reallocates terminal contributions while preserving complete port powers. A held-ring example transfers $987\pi/8$ joules at the sun in ground coordinates and $705\pi/8$ joules in carrier coordinates. A missing electrical return produces an exact apparent deficit of $21/4$ joules. Five compound geometries give frame-dependent maximum-power ratios from $100/441$ to $2950/21$, with unit ground-reference ratio throughout. Closed-form finite-state balances retain leakage, loading, preparation, switching, and nonzero final stores. A compliant electrical–mechanical correspondence is exact under its stated constitutive map, but a voltage coordinate shift does not physically realize accelerating-frame storage.

Latest PDF on GitHub:

<https://github.com/hobnilre/physics-gear/blob/main/frames-returns-and-port-power.pdf>

1 Observations and energy boundaries

A planet gear can turn relative to its carrier while an observer on the housing reports another angular displacement. A winding terminal can likewise be read against either of two returns. Neither observation specifies the work delivered to an attached receiver. Work requires the effort and flow at the receiver’s actual boundary, and attaching the receiver can change the motion or circuit. The purpose here is to separate these operations through exact derivations.

All angular components are positive by the right-hand rule about the common $+z$ axis. We use angular rates ω in radians per second and $\nu = \omega/(2\pi)$ in revolutions per second. A frame rotating at Ω reports $\omega^f = \omega - \Omega$. A port power is positive into its stated system. For an interval with event times t_e , define

$$W_j(t_0, t_1) = \int_{t_0}^{t_1} P_j(t) dt, \quad P_j = e_j f_j, \quad (1)$$

$$r_E = E(x(t_1)) - E(x(t_0)) - \sum_j W_j - \sum_e W_e^{\text{imp}}. \quad (2)$$

Here $e_j f_j$ means torque times angular rate, force dotted with velocity, or voltage times current; an exported heat channel can equivalently be written $-T\dot{S}_{\text{out}}$ for a reservoir at fixed temperature T . Each port is integrated separately before the sum is formed. A constitutive endpoint energy does not specify the work of an unmodelled preparation path.

We distinguish a whole-system boundary from component boundaries. A mesh or pin has two sides; they are indispensable when examining a planet but internal to an enclosure containing both connected bodies. Similarly, winding powers and terminal contributions must not be added again to the source, receiver, and heat powers of a complete circuit. Ideal constraints below have no energy coordinates; their empty store is zero by definition. Finite bodies and windings have explicit nonempty stores evaluated at both endpoints.

Signed sums also differ from absolute observations. For oriented coordinates y_j and positive weights w_j , $\sum_j w_j |y_j|$ depends on the observation map, units, and weights. A linear constraint on the signed coordinates does not make their absolute sum conserved, nor make it energy. For example, if $E = x^T H x / 2$ with positive definite H and $y = O x$, Cauchy–Schwarz gives the exact instantaneous geometric bound

$$\sum_j w_j |y_j| \leq \sqrt{2E(x) \max_{\sigma_j = \pm 1} (w \odot \sigma)^T O H^{-1} O^T (w \odot \sigma)}. \quad (3)$$

It uses the current state energy, not a presumption that the initial energy persists. Extra conserved linear functionals restrict the admissible set only when actually derived. Neither a gear turn count nor a raw winding-linkage sum is automatically eligible for a conserved-sum growth claim.

2 Rolling constraints and three kinds of count

2.1 A parallel-axis train

Let Z_s, Z_p, Z_r be positive integer tooth counts of the sun, each planet, and the internal ring. For common module m_0 , the pitch radii are $r_j = m_0 Z_j / 2$. The planet centre is at radius a . The simple train requires

$$a = r_s + r_p = r_r - r_p, \quad Z_r = Z_s + 2Z_p. \quad (4)$$

Pure rolling at the two pitch points gives two independently defined planet rates:

$$\omega_p = \omega_c - \frac{Z_s}{Z_p} (\omega_s - \omega_c), \quad (5)$$

$$\omega_p = \omega_c + \frac{Z_r}{Z_p} (\omega_r - \omega_c). \quad (6)$$

Eliminating ω_p yields the division-free Willis relation

$$Z_s (\omega_s - \omega_c) + Z_r (\omega_r - \omega_c) = 0. \quad (7)$$

The ratio form is valid only when its denominator is nonzero. Equation (7) also covers the locked train.

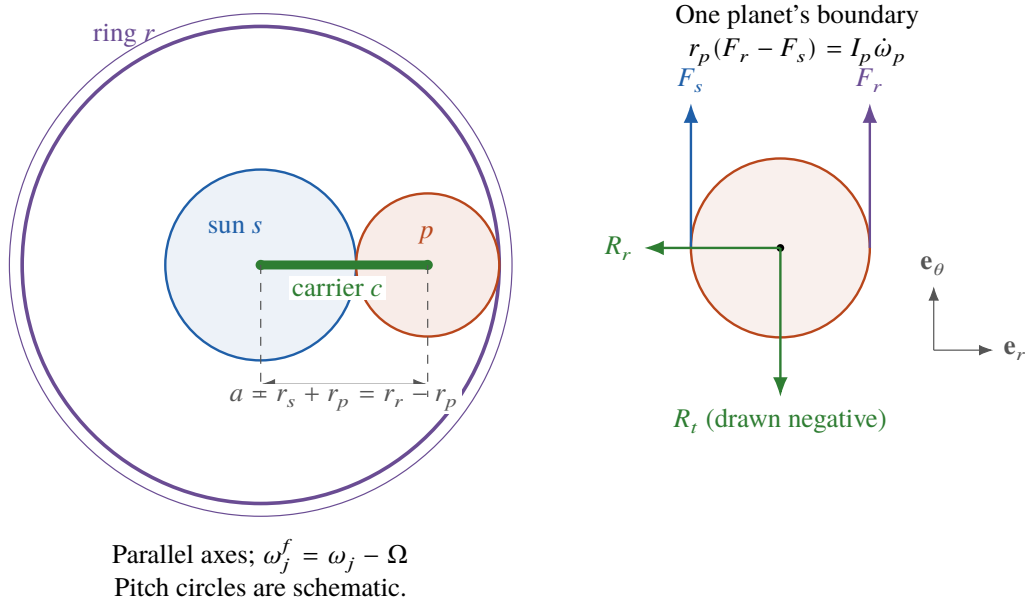


Figure 1. Schematic pitch geometry and forces on one planet. Blue identifies the sun, orange the planet, green the carrier, and purple the ring. Tangential mesh forces on the planet have signed components F_s, F_r ; its pin force is $R_t \mathbf{e}_\theta + R_r \mathbf{e}_r$. Only one of the three planets is drawn.

The geometry of Figure 1 permits scalar subtraction of rates because all rotation axes are parallel. Helical thrust, bevel geometry, and frames rotating about other axes require vector kinematics and additional force components. Positive pitch radii alone do not establish manufacturability: equally spaced multiple planets also require compatible tooth phasing and clearance. For the simple train with q equally spaced planets, a familiar phase condition follows by advancing adjacent mesh phases through $2\pi/q$: $(Z_s + Z_r)/q$ must be an integer. All algebra below can instead describe a single planet or a stipulated equal-load assembly; it is not proof that every integer pair admits three equally spaced, noninterfering planets.

2.2 Signed displacement, tooth passage, and a shaft readout

On the same explicit interval define

$$N_p^f = \frac{1}{2\pi} \int_{t_0}^{t_1} (\omega_p - \Omega) dt, \quad Q_p = \frac{Z_p}{2\pi} \int_{t_0}^{t_1} (\omega_p - \omega_c) dt. \quad (8)$$

Q_p is an oriented tooth-pitch passage at a named mesh; integer crossing counts additionally need an initial tooth phase. The sign at the other mesh can be chosen oppositely, but that choice must be retained. Absolute traffic is $Z_p \int |\omega_p - \omega_c| dt / (2\pi)$, which equals $|Q_p|$ only without a reversal. The magnitude of the net body count $|N_p^f|$ is another observable again.

Table 1. Three distinct meanings of a planet's accumulated rotation.

Observable	Definition	Dependence
Body displacement	N_p^f	Observation frame
Tooth-pitch passage	Q_p at a named mesh	Relative motion; invariant under a common frame change
Output displacement	$(\Delta\theta_o - \Delta\theta_\sigma)/(2\pi)$	Linkage and stator σ

Subtracting the integrands, without any dynamical assumption, proves

$$N_p^0 - N_p^c = N_c^0, \quad N_p^r = N_p^c + N_c^r. \quad (9)$$

A common frame subtraction leaves every relative rate unchanged and hence leaves Q_p unchanged. These identities apply to arbitrary integrable paths, including reversals; rational constant rates are only a convenient example.

For specificity, take the positive-integer family

$$Z_s \in \{12, 17, 18, 24, 30, 36, 48, 60, 101\}, \quad Z_p \in \{12, 13, 18, 24, 36\}.$$

It has $36 \leq Z_r \leq 173$ under (4). Over $[0, 7/2]$ s, eight distinct rate choices are described by the following complete definitions, with the other rates obtained from (6)–(7).

Table 2. Rate choices for ground, carrier, sun, and ring observations. The interval is not one second, so rates and accumulated counts must not be interchanged.

Choice	(ν_s, ν_c) in rev/s	Defining feature
1	$(7/3, 5/4)$	Forward motion
2	$(-2, 3)$	Counterrotation
3	$(1 + Z_r/Z_s, 1)$	Ring held
4	$(0, 1)$	Sun held
5	$(1, 0)$	Carrier held
6	$(1, 1)$	Locked relative motion
7	$(1 + Z_p/Z_s, 1)$	Planet stationary in ground
8	$(1, Z_s/(Z_s + Z_p))$	Same stationary condition with sun rate fixed

Write $d = \omega_s - \omega_c$ and $(k_s, k_c, k_r, k_p) = (1, 0, -Z_s/Z_r, -Z_s/Z_p)$. Every rate is $\omega_j = \omega_c + k_j d$, and every interbody relative rate is $(k_j - k_l)d$. The four k_j are distinct, so all relative rates vanish if and only if $d = 0$. In the full (ω_s, ω_c) plane, the four absolute-rate zeros and this relative-rate zero are **five lines** through the origin, not five rays. A transverse passage through a nontrivial linear zero reverses sign; a tangency or the identically zero path does not.

3 Forces, torques, and moving-frame work

3.1 Free bodies before power

Use $q = 3$ equally loaded planets, no pin couple, and inertias I_s, I_r, I_c, I_p ; m_p is one planet's mass. Mesh forces F_s, F_r are tangential components on one planet, with the signs in Figure 1. The following Newton–Euler equations determine the remaining forces and shaft torques from a prescribed τ_s and kinematically compatible accelerations:

$$F_s = \frac{\tau_s - I_s \dot{\omega}_s}{qr_s}, \quad F_r = F_s + \frac{I_p \dot{\omega}_p}{r_p}, \quad (10)$$

$$\tau_r = I_r \dot{\omega}_r + qr_r F_r, \quad R_t = m_p a \dot{\omega}_c - F_s - F_r, \quad (11)$$

$$R_r = -m_p a \dot{\omega}_c^2, \quad \tau_c = I_c \dot{\omega}_c + qaR_t. \quad (12)$$

The radial equation assumes fixed a and neglects other radial mesh forces; it isolates the declared tangential pitch-contact model. A pressure-angle model would change the radial reaction.

When all rates are constant, (12) gives

$$F_r = F_s = \frac{\tau_s}{qr_s}, \quad \tau_r = \frac{Z_r}{Z_s} \tau_s, \quad \tau_c = -\frac{Z_s + Z_r}{Z_s} \tau_s, \quad \tau_s + \tau_r + \tau_c = 0. \quad (13)$$

These are consequences of the free bodies. The planet's real axial couple $r_p(F_r - F_s) = I_p \dot{\omega}_p$ vanishes exactly when $\dot{\omega}_c = (Z_s/Z_p)(\dot{\omega}_s - \dot{\omega}_c)$ for $I_p > 0$. Thus a steady idler has zero net spin-couple work in either frame, although its body count differs. Its separate contact and pin powers need not vanish.

Let

$$J = I_s + I_r + I_c + q(I_p + m_p a^2), \quad (14)$$

$$H = I_s \omega_s + I_r \omega_r + (I_c + qm_p a^2) \omega_c + qI_p \omega_p = J \omega_c + Bd, \quad (15)$$

$$B = I_s - I_r Z_s/Z_r - qI_p Z_s/Z_p. \quad (16)$$

Summing the free bodies gives $\sum \tau = \dot{H}$ on (4). Before imposing that assembly, the surplus is $q[(r_s - a + r_p)F_s + (r_r - a - r_p)F_r]$. It need not vanish. This is a compatibility requirement of these equations, not a failure of the angular-momentum law for physical mechanisms.

The equation $\sum \tau = 0$ alone constrains accelerations to $J\dot{\omega}_c + Bd = 0$ and need not make them zero. For example, choose positive effective inertias $I_s = I_r = qI_p = I_c + qm_p a^2 = 1 \text{ kg m}^2$ and tooth ratios $Z_s/Z_r = 2/5$, $Z_s/Z_p = 4/3$. Then $J = 4$, $B = -11/15$ in the corresponding units. Taking $\dot{\omega}_c = 1 \text{ rad/s}^2$ and $\dot{d} = 60/11 \text{ rad/s}^2$ accelerates every body while $\dot{H} = 0$. The torque ratios in (13) cannot be inferred in this case.

3.2 The actual port sides

The train boundary encloses all gears, carrier, and pins. Its physical inputs are the three shafts, including the held-member reaction. Internal transfers on one planet's boundary are listed separately:

Table 3. Force–velocity and torque–rate products. Here $\dot{a} = 0$. Rolling equates the two contact-point velocities; it does not set each contact-side power to zero.

Port or side	Signed instantaneous power in frame f
Shaft $j = s, r, c$ into complete train	$\tau_j(\omega_j - \Omega)$
Sun mesh into one planet	$F_s r_s(\omega_s - \Omega)$
Same mesh into sun	$-F_s r_s(\omega_s - \Omega)$
Ring mesh into one planet	$F_r r_r(\omega_r - \Omega)$
Same mesh into ring	$-F_r r_r(\omega_r - \Omega)$
Pin into one planet	$R_t a(\omega_c - \Omega) + R_r \dot{a}$
Same pin into carrier	Negative of planet-side pin power

A sliding variant must replace ideal no-slip contact by a declared relative velocity v_{slip} and opposing friction force. With Coulomb coefficient $\mu_f \geq 0$ and normal force $F_N \geq 0$, the pair's mechanical power sum is $-D$, where $D = \mu_f F_N |v_{\text{slip}}|$. Both contact points receive the same subtraction $\Omega \times \mathbf{r}$, so v_{slip} and D are frame invariant. The exported heat work is $-\int D dt$, or $-\int T \dot{S}_{\text{out}} dt$ under immediate thermal export. The pure-rolling idealization has $D = 0$; it does not bound real mesh losses, even for an illustrative $0 \leq \mu_f \leq 1/2$.

3.3 Relative kinetic energy and effective centrifugal potential

Define the constitutive ground kinetic energy

$$K^0 = \frac{1}{2} I_s \omega_s^2 + \frac{1}{2} I_r \omega_r^2 + \frac{1}{2} (I_c + q m_p a^2) \omega_c^2 + \frac{1}{2} q I_p \omega_p^2. \quad (17)$$

Replacing every angular rate by its frame-relative value gives $K^f = K^0 - \Omega H + J \Omega^2/2$. The effective centrifugal potential of all the mass, including spin contributions, is $U^f = -J \Omega^2/2$. Hence

$$E^f = K^f + U^f = K^0 - \Omega H, \quad E^0 - E^f = \Omega H. \quad (18)$$

For one planet the centrifugal part is $-\Omega^2 (I_p + m_p a^2)/2$, of which the orbital part is $-m_p \Omega^2 a^2/2$. This is a coordinate-dependent effective potential, not a newly created physical reservoir. It can make E^f negative. For fixed J , the Euler-force power summed over the bodies is $-\dot{\Omega}(H - J\Omega)$: for a spin it is the fictitious torque $-I\dot{\Omega}$ times $\omega - \Omega$, and for translation it is $(-m\dot{\Omega} \times \mathbf{r}) \cdot \mathbf{v}^f$. The explicit potential-time term is $-J\Omega\dot{\Omega}$. They must be retained separately when using E^f . The second is a parameter term in the effective store, not a physical shaft or a second heat loss. Multiplying (12) by the corresponding velocities gives

$$\dot{K}^0 = \sum_{j=s,r,c} \tau_j \omega_j - D, \quad (19)$$

$$\dot{E}^f = \sum_{j=s,r,c} \tau_j (\omega_j - \Omega) - D - \dot{\Omega}(H - J\Omega) - J\Omega\dot{\Omega}. \quad (20)$$

Thus a complete effective-frame balance includes $-\dot{\Omega}H$ in addition to physical port products. No physical frame-driving supply work is inferred from this mathematical term.

A useful correction concerns Coriolis force. At fixed a the planet-centre velocity is $a(\omega_c - \Omega)\mathbf{e}_\theta$ in frame f , so

$$\mathbf{F}_{\text{Cor}} = 2m_p \Omega a(\omega_c - \Omega)\mathbf{e}_r, \quad \mathbf{F}_{\text{Cor}} \cdot \mathbf{v}^f = 0. \quad (21)$$

It is zero in ground and carrier frames, but generally nonzero in a sun or ring frame. Constant radius guarantees zero Coriolis **power**, not zero Coriolis force in every frame. Radial motion adds a tangential component; Coriolis force still does no total instantaneous work.

The real axial torque component is unchanged by rotating coordinates about z , even during acceleration. Real force vectors are also unchanged as geometrical objects, but their coordinate components are not. If R_r, R_t are constant in the carrier basis, then $R_x = R_r \cos \phi - R_t \sin \phi$ and $R_y = R_r \sin \phi + R_t \cos \phi$, with $\dot{\phi} = \omega_c - \Omega$. Each fixed-axis component has full-cycle peak-to-peak excursion $2\sqrt{R_r^2 + R_t^2}$; a shorter interval need not reach those extrema. Stationarity requires zero relative rotation or zero force, with additional possibilities only for specially varying components.

3.4 An exact steady ledger and its preparation

Choose $(Z_s, Z_p, Z_r) = (24, 18, 60)$, $m_0 = 1/500$ m, $\tau_s = 47/4$ N m, and held-ring rates $(\nu_s, \nu_c, \nu_r, \nu_p) = (7/2, 1, 0, -7/3)$ rev/s. The operation interval is $[2/5, 19/10]$ s, of duration $3/2$ s. Each work in the next table is its constant effort–flow product times that duration, integrated before combining.

Table 4. Exact operation works. Opposite internal sides have opposite signs and cancel on the whole-train boundary. The sun and ring mesh sides are different; there is no single work common to every mesh side.

Signed work	Ground, J	Carrier, J
Sun shaft	$987\pi/8$	$705\pi/8$
Held-ring shaft	0	$-705\pi/8$
Carrier shaft	$-987\pi/8$	0
One planet-side pin	$-329\pi/8$	0
One sun-mesh planet side	$329\pi/8$	$235\pi/8$
One ring-mesh planet side	0	$-235\pi/8$

For endpoint evaluation choose $m = \mu r^2$, $I = \mu r^4/2$ for each disc, with $\mu = 490$ kg/m²; μ includes the circular area factor. The carrier disc has radius $a = 21/500$ m. The ring has outer radius $r_o = r_r + 1/250$ m and $I_r = \mu(r_o^4 - r_r^4)/2$. Planet orbital inertia is additional. Direct substitution gives

$$J = \frac{16851149}{6250000000} \text{ kg m}^2, \quad H_* = \frac{166698\pi}{48828125} \text{ kg m}^2/\text{s}, \quad (22)$$

$$K_*^0 = \frac{18864657\pi^2}{3125000000} \text{ J}, \quad E_*^c = -\frac{2472687\pi^2}{3125000000} \text{ J}. \quad (23)$$

Both endpoints of operation have these respective values because their states are explicitly the same. The separately summed external shaft works are zero in either frame, and $r_E = 0$. A held ring has nonzero carrier-frame work; deleting its reaction port would leave $705\pi/8$ J unbalanced. This difference is present in shaft and mesh sides as well as pins.

The same endpoint can be reached by a specified smooth preparation. Put $F(z) = 3z^2 - 2z^3$; on $[0, 2/5]$ s take $\omega_j = \omega_{j*}F(t/(2/5))$ and $\tau_s = (47/4)F$. On $[19/10, 5/2]$ s replace F by $1 - F((t - 19/10)/(3/5))$. Accelerations and applied torque meet the steady values continuously. There are no impulses. The free bodies yield $\tau_j = a_j F + b_j \dot{F}$, where

$$(a_s, a_r, a_c) = \left(\frac{47}{4}, \frac{235}{8}, -\frac{329}{8}\right), \quad (b_s, b_r, b_c) = \left(0, -\frac{1639197\pi}{625000000}, \frac{18864657\pi}{3125000000}\right) \quad (24)$$

in torque and angular-momentum units respectively. Each shaft integral on a ramp of duration T is exactly

$$W_j^f = (\omega_{j*} - \Omega_*) \left(\frac{13}{35} T a_j \pm \frac{1}{2} b_j \right), \quad (25)$$

with $+$ for preparation and $-$ for relaxation. This follows from $\int_0^1 F^2 dz = 13/35$ and $\int F dF = \pm 1/2$. For the internal sides write

$$F_s = f_s F + g_s \dot{F}, \quad F_r = f_r F + g_r \dot{F}, \quad R_t = -2f_s F + g_t \dot{F},$$

where $f_s = (47/4)/(qr_s)$, $g_s = -I_s \omega_{s*}/(qr_s)$, $g_r = g_s + I_p \omega_{p*}/r_p$, and $g_t = m_p a \omega_{c*} - g_s - g_r$. The separately integrated planet-side works are therefore

$$W_{sp}^f = r_s (\omega_{s*} - \Omega_*) \left(\frac{13T}{35} f_s \pm \frac{g_s}{2} \right), \quad (26)$$

$$W_{rp}^f = r_r (\omega_{r*} - \Omega_*) \left(\frac{13T}{35} f_r \pm \frac{g_r}{2} \right), \quad (27)$$

$$W_{pin,p}^f = a (\omega_{c*} - \Omega_*) \left(-\frac{26T}{35} f_s \pm \frac{g_t}{2} \right). \quad (28)$$

Each opposite side has the negative of its corresponding expression; no internal sum is added to the whole-train shaft works. The ground endpoints are $0, K_*^0$ during preparation and $K_*^0, 0$ during relaxation. Carrier endpoints are $0, E_*^c$ and $E_*^c, 0$.

The separately integrated Euler and potential-time terms on preparation are $-\Omega_*(H_* - J\Omega_*)/2$ and $-J\Omega_*^2/2$; both reverse on relaxation. The ground-minus-carrier shaft-work sum is $\Omega_* H_*/2 = 166698\pi^2/48828125$ J, and the corresponding effective-field difference is the same. Their total is $\Omega_* H_* = 333396\pi^2/48828125$ J. For the distinct forward rates $(\nu_s, \nu_c) = (7/3, 5/4)$ the same formula applies with its own H_* ; a carrier-held path has $\Omega_* = 0$ and no frame difference. These statements follow from the path, not constant stores.

In general the shaft difference is $\int \Omega dH$ and the effective-field difference is $\int H d\Omega$. Integration by parts proves

$$\int_{t_0}^{t_1} \Omega \dot{H} dt + \int_{t_0}^{t_1} H \dot{\Omega} dt = [\Omega H]_{t_0}^{t_1}. \quad (29)$$

The equal split requires a proportional history. For the explicit path $H = H_* z$, $\Omega = \Omega_* [z + bz(1 - z)]$, $0 \leq z \leq 1$, its two shares are $1/2 + b/6$ and $1/2 - b/6$. They can lie outside $[0, 1]$ when one term opposes the total. When $B \neq 0$ in (16), compatible sun and carrier rates can be reconstructed from H and $\Omega = \omega_c$. Prescribing only endpoint rates leaves these individual works undetermined.

3.5 The locked control and the two limits

Set every rate to 2π rad/s and set the drive torque to zero. Equation (12) leaves only $R_r = -83349\pi^2/3125000$ N, with zero radial power. Over $[0, 3/2]$ s, $N_p^0 = N_c^0 = 3/2$, $N_p^c = 0$, and no teeth pass either mesh. Every physical port work is exactly zero. Independent stores are

$$K^0(t_0) = K^0(t_1) = \frac{16851149\pi^2}{3125000000} \text{ J}, \quad E^c(t_0) = E^c(t_1) = -\frac{16851149\pi^2}{3125000000} \text{ J}. \quad (30)$$

In carrier coordinates $K^c = 0$, whereas U^c is negative. Assigning the positive ground value to both effective stores would contradict (18). Both balances nonetheless give $r_E = 0$.

Locking the rates while retaining $\tau_s = 47/4 \text{ N m}$ is a different control. It transmits the self-equilibrated torque set (13); for example the ground pin power per planet is $-329\pi/12 \text{ W}$ and its carrier-frame power is zero. On the approach $\omega_c = 2\pi$, $\omega_s = \omega_c + \varepsilon$, fixed drive gives a nonzero limiting maximum shaft work $987\pi/8 \text{ J}$ over $3/2 \text{ s}$; that maximum is constant on the portion where the carrier port dominates. Scaling the drive to zero with ε makes every port work approach zero. Both paths are continuous, but only the latter reaches the zero-drive null. The varying counts alone imply neither result about work.

4 What a physical lead-out reports

4.1 Stator, phase reference, and finite-interval ripple

A lead-out connects the planet axle to a parallel output axis offset by a ; it does not connect that output to the sun. Denote the planet and carrier angles by θ_p, ϕ , and the output and stator angles by θ_o, θ_σ . The reported count is $N_o^\sigma = \Delta(\theta_o - \theta_\sigma)/(2\pi)$. For the **same output motion**, subtraction gives

$$N_o^0 - N_o^c = \Delta\phi/(2\pi). \quad (31)$$

With finite loading, physically remounting the stator may change that motion; then (31) must be applied separately to each trajectory.

For a Hooke joint of angle $\beta < \pi/2$, let $c_\beta = \cos \beta > 0$ and let $H_\beta(x)$ be the continuous lift of $\text{atan2}(\sin x, c_\beta \cos x)$, with $H_\beta(0) = 0$. Then $H_\beta(x + \pi) = H_\beta(x) + \pi$ and $\tan H_\beta(x) = \tan x/c_\beta$. The joint plane in a parallel-offset Z arrangement is carried by the carrier. Writing $x = \theta_p - \phi$, a declared two-joint kinematic map is

$$\theta_o = \phi + H_\beta\left(H_\beta(x) + \gamma + \frac{\pi}{2}\right) + \frac{\pi}{2}, \quad (32)$$

where constant additive angles have no effect on displacement. With correct carrier-fixed phasing $\gamma = 0$, this reduces to $\theta_o = \theta_p$ up to a constant. A rigid intermediate shaft supplies that fixed relative phasing. With carrier-fixed quarter-turn error $\gamma = \pi/2$, write the output as $\phi + F_\beta(x)$ up to a constant; then

$$\tan F_\beta(x) = \frac{\tan x}{c_\beta^2}, \quad q_\beta(x) = F'_\beta(x) = \frac{c_\beta^2}{c_\beta^4 \cos^2 x + \sin^2 x}, \quad c_\beta^2 \leq q_\beta \leq c_\beta^{-2}. \quad (33)$$

This is a twice-per-relative-revolution ripple. Since $F_\beta(x) - x$ is bounded and π -periodic, the mean count agrees with the planet count, but an arbitrary finite interval has the endpoint correction $\delta = [F_\beta(x) - x]_{t_0}^{t_1}/(2\pi)$. It is exactly zero over complete ripple periods. It must not be dropped merely because the mean ratio is one.

Holding the phase in ground instead prescribes $\gamma = -\phi$. Equation (32) then gives $\theta_o = \theta_p - \phi +$ a bounded phase-dependent term. A grounded stator has the carrier-relative **mean** count, with an endpoint ripple correction δ_g determined by that equation. An exact claim for all finite intervals would be false. The device that enforces this drifting phase has not been specified dynamically; its required effort and supply work remain unknown.

Table 5. Eleven mountings, with finite-interval phase corrections retained. Coincidence with a particular body count in a special state is not an identity for all states.

Coupling and phase	Ground stator count	Carrier stator count
Correct double Hooke, carrier phase	N_p^0	N_p^c
Quarter-turn double Hooke, carrier phase	$N_p^0 + \delta$	$N_p^c + \delta$
Double Hooke, ground phase	$N_p^c + \delta_g$	Not assigned a physical mounting here
Ideal Oldham or equivalent constant-ratio offset path	N_p^0	N_p^c
Bevel path, $\kappa = 1$	N_p^0	N_p^c
Bevel path, $\kappa = 2$	$N_c^0 + 2N_p^c$	$2N_p^c$

The ideal Oldham constraint is $\theta_o = \theta_p$ even as the offset precesses. Orthogonal sliding coordinates are $a \cos x$ and $a \sin x$; each has excursion amplitude a (full stroke $2a$), and their maximum absolute value never exceeds a . This kinematic statement can also represent an ideal constant-velocity Schmidt path; it does not make their detailed force systems identical. The bevel path obeys $\theta_o = \phi + \kappa(\theta_p - \phi)$, which proves the last two cases.

Whether an actual orbiting lead-out follows these phase and stator predictions under a small load remains a measurement question [OP-EPI-05]. A carrier encoder and an output encoder can resolve the predicted full-turn count and the twice-per-relative-revolution ripple at low speed; changing the phase datum distinguishes a carrier-fixed joint plane from a ground-referenced actuator. Their finite-interval differences must include δ or δ_g . These are ordinary angle observations, without exceptional energy sensitivity. Equation (31) rules out a single unchanged rotor angle having equal readings against two differently rotating stators. It does not prove that no differential device can encode relative tooth passage, or that no device of any kind can produce an invariant output.

4.2 Load, support, and cycle work

For the carrier-fixed map $\theta_o = \phi + F(x)$ and constant input rates, put $d = \omega_p - \omega_c$, $q = F'(x)$, and $\omega_o = \omega_c + qd$. A linkage boundary includes a rotor of spin inertia I_R and mass m_R orbiting at a , an output inertia I_o , and massless coupling members. Its ground kinetic store is $K_L^0 = I_R \omega_p^2/2 + m_R a^2 \omega_c^2/2 + I_o \omega_o^2/2$; its other-frame stores follow (18) with their own J, H . The stator, imposed output torque τ_L , and drive are external.

Let the drag magnitude be $d_0 \geq 0$, with $s_\sigma = \text{sgn}(\omega_o - \omega_\sigma)$ away from a zero. The output equation gives coupling torque $T = I_o \dot{\omega}_o - \tau_L + d_0 s_\sigma$. Virtual angular displacements in $\theta_o(\theta_p, \phi)$ give $\tau_d = Tq$ and $\tau_u = T(1 - q)$ at the drive and carrier support. Their work integrals and those of the load, drag, and rotor pin are

$$W_d^f = \int Tq(\omega_p - \Omega)dt, \quad W_u^f = \int T(1 - q)(\omega_c - \Omega)dt, \quad (34)$$

$$W_L^f = \int \tau_L(\omega_o - \Omega)dt, \quad W_{\text{drag}}^f = \int (-d_0 s_\sigma)(\omega_o - \Omega)dt, \quad (35)$$

$$W_{\text{pin}}^f = \int m_R a \dot{\omega}_c a(\omega_c - \Omega)dt. \quad (36)$$

All integrals in this section have the explicitly stated readout or cycle interval. At constant ω_c the rotor-

pin power is zero, whereas the support port can be nonzero. In accelerating motion the pin and effective frame terms must both be restored. Summing the first four instantaneous products yields $I_o \dot{\omega}_o (\omega_o - \Omega)$; the rotor and orbital terms complete the independently evaluated store change. This torque-path model does not resolve transverse Cardan joint forces or bending-couple ports; two joints alone do not determine that system. Attaching it to a train with unchanged prescribed torques is not a solved combined loaded apparatus.

The physical dissipation is $D_\sigma = d_0 |\omega_o - \omega_\sigma|$. It equals minus the rotor drag power only in the stator frame. For the reference train's ideal Oldham path, $d_0 = 1/50$ N m and $[0, 7/2]$ s give

$$\int D_0 dt = \frac{49\pi}{150} \text{ J}, \quad \int D_c dt = \frac{7\pi}{15} \text{ J}. \quad (37)$$

These are different physical mountings. A change of coordinates on either fixed mounting leaves its relative drag dissipation unchanged. A prescribed $\tau_L = -47/40$ N m is not necessarily a passive load: when $\omega_o < 0$ its ground-frame power is positive. Describing a torque as a brake therefore requires its relative velocity as well as its sign.

For the quarter-turn Hooke path, one cycle has $T_b = \pi/|d|$. Assume τ_L and s_σ remain constant through that cycle. Set $T_0 = -\tau_L + d_0 s_\sigma$. Since $F(x + \pi) - F(x) = \pi$, $\int (1 - q) dt = 0$. Also $\dot{\omega}_o = d\dot{q}$. The ground support work integrates exactly to

$$W_u^0 = \omega_c \left\{ T_0 \int (1 - q) dt + I_o d \left[q - \frac{q^2}{2} \right]_{t_0}^{t_0 + T_b} \right\} = 0. \quad (38)$$

The output rate and all stores return to their initial values. This proves zero cycle support work for this torque path; it makes no claim for changing drag sign, compliance, backlash, joint friction, or the unresolved transverse force system. It is stronger within that model than merely observing a small cycle residual, and narrower than a claim about real joints.

For an axial lead-out length $L = 3/25$ m, the geometry gives $c_\beta^2 = L^2/(L^2 + a^2)$. Choosing $a = 137/1000$ m gives $c_\beta^2 = 14400/33169$, the exact illustrative value in Figure 2. Its large articulation is a kinematic example only.

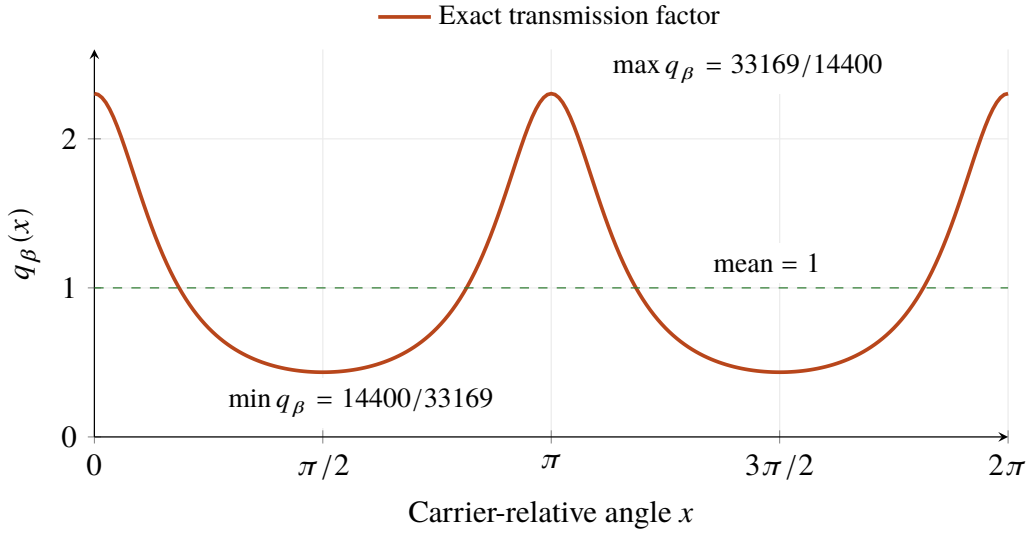


Figure 2. Exact quarter-turn Hooke transmission factor from equation (33), with $\cos^2 \beta = 14400/33169$. The ripple is periodic in the carrier-relative input angle. Its mean is one even though the instantaneous factor ranges between the annotated exact bounds. The illustrative articulation is not asserted to be practical.

For the bevel path $q = \kappa$, the support work is instead $T_0(1 - \kappa)(\omega_c - \Omega)T_b$. As an exact example choose $Z_s = Z_p = 12$, $Z_r = 36$, $(v_s, v_c) = (7/3, 5/4)$, $\kappa = 2$, $\tau_L = -47/40$, $d_0 = 1/50$ in SI units. Then $v_p = 1/6$, $v_o = -11/12$, $T_b = 6/13$ s, and $T_0 = 231/200$ N m for either listed stator. Thus

$$W_u^0 = -\frac{693\pi}{520} \text{ J}, \quad W_u^c = 0. \quad (39)$$

The other integrals in (36) balance it with unchanged endpoint stores. Nonzero work at one support is a transfer across that boundary; it is not a residual or an independent energy source.

5 Windings, returns, and the electrical observation

5.1 The tapped and isolated connections

Winding 1 is oriented from terminal a to common return c ; winding 2 is oriented from b to a . Current enters each winding at its first terminal. Define

$$v_1 = V_a - V_c, \quad v_2 = V_b - V_a, \quad v_o = v_1 + v_2, \quad (I_a, I_b, I_c) = (i_1 - i_2, i_2, -i_1). \quad (40)$$

The winding-assembly currents obey $I_a + I_b + I_c = 0$. A source supplying an extra shunt resistor has another current I_s , not simply I_a . Figure 3 specifies the conductive connections independently of winding polarity and magnetic coupling.

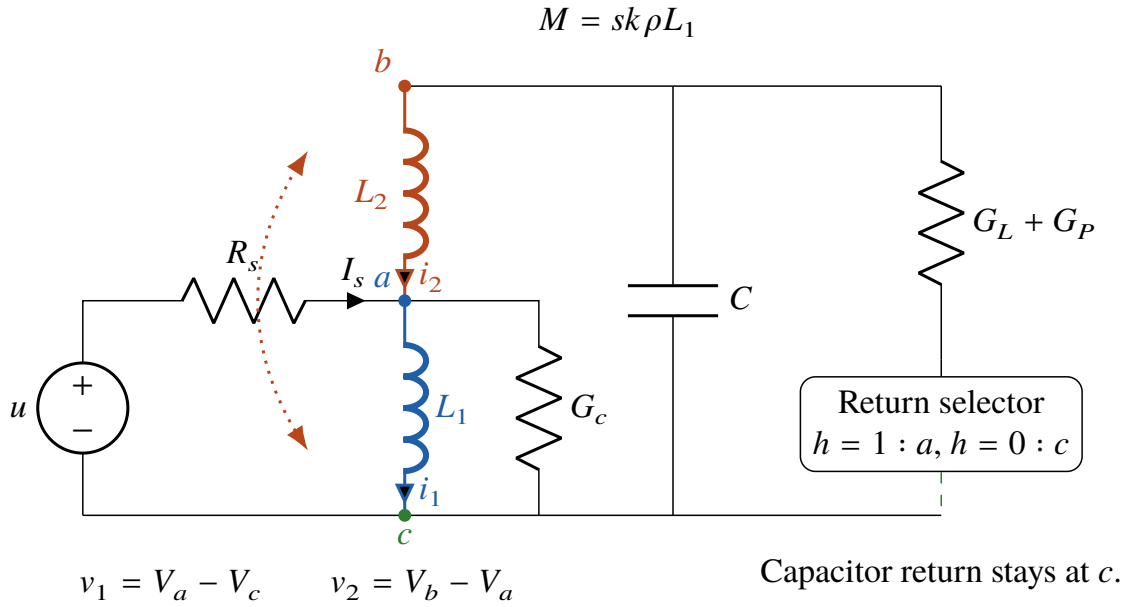


Figure 3. Schematic tapped circuit. The source reaches a through R_s ; the coupled winding sections are $a-c$ and $b-a$. The capacitor remains across $b-c$. The receiver and probe may return to a or c , while the effective core-loss conductance remains across $a-c$. Dotted coupling is magnetic, not a wire.

An isolated control keeps winding 1 across its source and places winding 2 across its own capacitor and receiver. Its two returns are physically distinct; assigning coordinate potential zero to each does not connect them. When terminal contributions are combined, both returns must be retained.

The standard distinction between a finite coupled-inductance model and an ideal transformer is explicit in [Haus and Melcher \(1989\)](#), Section 9.7. Here we use it as a constitutive distinction and derive the required power relations directly. Put $N_1, N_2 > 0$, polarity $s = \pm 1$, and $n = sN_2/N_1$. A shared-flux idealization has $\lambda_1 = N_1 \Phi$, $\lambda_2 = sN_2 \Phi$. Its voltage and ampere-turn constraints are

$$v_2 = nv_1, \quad i_1 = -ni_2, \quad v_o = (1+n)v_1, \quad I_a = -(1+n)i_2. \quad (41)$$

The first follows by differentiation of the linkage relations, and the second from $N_1 i_1 + sN_2 i_2 = 0$. They imply zero total ideal winding power without using power balance to define the currents. Three terminals give two independent voltages, not three independent two-terminal energy ports.

The weighted linkage constraint is $\lambda_2 - n\lambda_1 = 0$, whereas

$$|\lambda_1| + |\lambda_2| = (1 + |n|)|\lambda_1|, \quad \lambda_1 + \lambda_2 = (1 + n)\lambda_1. \quad (42)$$

The absolute sum changes with turns ratio, and the unweighted signed sum usually changes with excitation. Finite leakage further prevents reducing both linkages to one core flux. These observations do not establish an unweighted conserved-state functional or an absolute-sum growth envelope.

5.2 Which terminal represents which body

For the restricted ring-mesh correspondence choose planet $\leftrightarrow b$, carrier $\leftrightarrow a$, and ring $\leftrightarrow c$. Then

$$\omega_p - \omega_c = -\frac{Z_r}{Z_p}(\omega_c - \omega_r) \quad \leftrightarrow \quad v_2 = nv_1, \quad n = -Z_r/Z_p < -2. \quad (43)$$

The last inequality uses positive Z_s, Z_p and (4). Positive ratios and the opposed null $n = -1$ are useful electrical controls outside this simple planetary assignment.

Define a signed voltage integral, rather than a copper-turn count,

$$\Psi_{xy}(t_0, t_1) = \int_{t_0}^{t_1} (V_x - V_y)dt, \quad \Psi_{bc} = \Psi_{ba} + \Psi_{ac}. \quad (44)$$

It has volt-second units, not joules. For $(Z_s, Z_p, Z_r) = (24, 12, 48)$ with ring held and carrier at 1 rev/s over $[0, 1]$ s, $N_p^c = -4$ and $N_p^r = -3$. Under the kinematic scale 1 V/(rev/s), $v_1 = 1$ V, $n = -4$, $\Psi_{ba} = -4$ V s, and $\Psi_{bc} = -3$ V s. Equation (43) proves the corresponding identity for every rate choice in the earlier family, not only this example.

An energy correspondence needs a dimensional scale α using radians:

$$V = \alpha\omega, \quad I = \tau/\alpha, \quad VI = \tau\omega, \quad C = J/\alpha^2. \quad (45)$$

Thus angular speed corresponds here to voltage, torque to current, and inertia to capacitance. The finite magnetic store below is not automatically an inertia store. A voltage integral also needs its initial linkage and copper correction:

$$\Delta\lambda_j = \int_{t_0}^{t_1} (v_j - R_j i_j)dt. \quad (46)$$

For the series readout both linkage changes and both copper drops contribute. No single node-potential integral is identified with core flux. An independent sun terminal, accelerating reference supply, and independently phased Hooke linkage are not realized by the three-terminal kinematic map.

5.3 A common zero and a missing return

For a fixed physical circuit, changing every potential to $V_j + g(t)$ leaves all complete port voltages unchanged. Individual terminal contributions $q_j = V_j I_j$ instead obey

$$q_j^g - q_j = g I_j, \quad \sum_j (q_j^g - q_j) = g \sum_j I_j = 0, \quad W_j^g - W_j = \int_{t_0}^{t_1} g I_j dt. \quad (47)$$

The identity includes the return, even when its original potential is zero. It holds for time-dependent g as well as a constant. Load power is $(V_b - V_c)i_L$, not one terminal contribution $V_b i_L$.

Take the ideal winding assembly as boundary, with $n = 1$, $G_L = 1$ S, and $v_1 = (t - 2)$ V for time t in seconds on $[0, 7/2]$ s. There are no topology changes or impulses; the voltage passes continuously through zero at 2 s. Writing $v = t - 2$ numerically in SI units, the currents are $(I_a, I_b, I_c) = (4v, -2v, -2v)$ and baseline potentials $(v, 2v, 0)$. The two primitives are $\int v^2 dt = 91/24$ and $\int v dt = -7/8$. Integrating each signed terminal product gives:

Table 6. Exact missing-return control on the ideal winding boundary. The physical receiver obtains $91/6$ J in either coordinate system.

Signed work or balance	$V_c = 0, \text{ J}$	$g = 3 \text{ V, J}$
W_a	$91/6$	$14/3$
W_b	$-91/6$	$-119/12$
W_c	0	$21/4$
Sum of complete terminal works	0	0
Initial and final ideal stores	$0, 0$	$0, 0$
r_E	0	0

Omitting c after the shift yields $r_E = +21/4$ J. It is exactly the missing return work, determined by its own integral rather than assigned from the residual. The same polynomial proof covers $n \in \{-4, -2, -101/100, -1, -99/100, -1/2, -1/4, 1/4, 1/2, 1, 2, 4\}$ and $g = g_0 + g_1 t$, including $g_0 \in \{-3, 0, 3\} \text{ V}$, $g_1 \in \{-1, 0, 1\} \text{ V/s}$. It also covers nonpolynomial common offsets such as $g = 3 + (7/10) \cos(5t) \text{ V}$ by (47). The ideal endpoint store is an empty sum, not a claim that a real core contains no energy.

Physical grounding remains different: a real bond, probe common-mode branch, or chassis capacitance adds a path and potentially a store [OP-TRF-01]. A low-voltage isolated winding, known shunt conductance G_b , and ordinary differential voltage and current readings can distinguish an added dissipative return from a mere coordinate shift. If the attachment is held at constant voltage difference V_b for duration T , its separate heat delivery is $G_b V_b^2 T$ exactly; without that held-voltage assumption the measured instantaneous product must be integrated on its actual path. Floating and bonded arrangements can therefore be distinguished at ordinary voltage and current levels. Stray capacitance adds $C_b V_b^2 / 2$ endpoint storage; its unknown magnitude is not determined by the gauge identity.

6 Loaded readouts and finite stores

6.1 Two exact receiving experiments

Moving a negligible-loading probe return from a to c selects a different voltage on one trajectory. Moving a finite load changes the circuit; the two trajectories must be established independently. Equation (44) holds within each trajectory, not across two different loaded experiments.

Use the ideal algebraic boundary with $n = -4$, a source fixing $v_1 = 1 \text{ V}$, and a 1 S load on $[0, 1] \text{ s}$. For return a , the load current is $j_L = -4 \text{ A}$ and source current is 16 A ; for return c they are -3 A and 9 A . The load-side currents into the transformer are $-j_L$ at b and $+j_L$ at the return. Direct integration gives:

Table 7. Separate exact experiments, with no internal event or impulse. The two load contributions decompose the load work and are not added again.

Quantity	Return <i>a</i>	Return <i>c</i>
Loaded voltage integral, V s	−4	−3
Source work into boundary, J	16	9
Complete load work into boundary, J	−16	−9
Load contribution at <i>b</i> , J	−12	−9
Load contribution at its return, J	−4	0
Initial/final ideal store, J	0/0	0/0
r_E , J	0	0

The extra 7 J of load delivery accompanies an independently integrated extra 7 J of source input. If instead $v_1 = (t - 2)$ V on $[0, 4]$ s, all three signed voltage integrals are zero, while

$$W_{L,a} = -16 \int_0^4 (t - 2)^2 dt = -\frac{256}{3} \text{ J}, \quad W_{L,c} = -9 \int_0^4 (t - 2)^2 dt = -48 \text{ J}. \quad (48)$$

The source products integrate separately to the opposite values. A continuous sign reversal changes neither the ideal empty endpoints nor the balance. Zero net readout is not zero transferred work, as Figure 4 illustrates.

The complete curves on $0 \leq t \leq 4$ s, in the stated SI units, follow from the polynomial primitives:

$$\Psi_{ba}(0, t) = -4(t^2/2 - 2t), \quad \Psi_{bc}(0, t) = -3(t^2/2 - 2t), \quad (49)$$

$$-W_{L,a}(0, t) = 16(t^3/3 - 2t^2 + 4t), \quad -W_{L,c}(0, t) = 9(t^3/3 - 2t^2 + 4t). \quad (50)$$

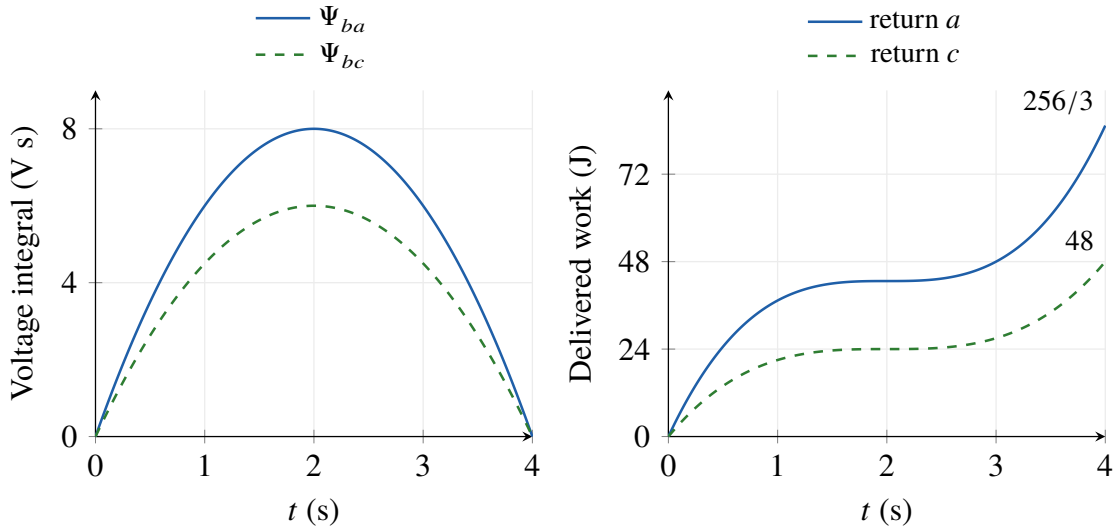


Figure 4. Exact reversing-drive control: the voltage integrals return to zero, while delivered load works increase as the integral of squared voltage. Time is in seconds; the two load connections are distinct ideal experiments driven by the same prescribed $v_1 = t - 2$.

For a general ideal ratio and a b – c resistive load, $i_2 = -G_L(1+n)v_1$, $i_1 = nG_L(1+n)v_1$ and $P_L = -G_L(1+n)^2v_1^2$. At $n = -1$ the output is identically zero; without imposed circulating current all terminal currents vanish while both winding voltages can be nonzero. For $n = -1 + \epsilon$, load work is $-G_L\epsilon^2 \int v_1^2 dt$ and approaches zero continuously, while winding-power direction reverses across the null. The values $-101/100, -1, -99/100$ are explicit two-sided controls. The b – a load instead sees nv_1 and need not vanish there. A finite transformer does not automatically inherit this ideal null.

6.2 The regular coupled-winding model

Let $\rho = N_2/N_1 > 0$, $s = \pm 1$, and $0 \leq k < 1$. Define

$$L_2 = \rho^2 L_1, \quad M = sk\rho L_1, \quad \mathbf{L} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix}, \quad E_m = \frac{1}{2} i^T \mathbf{L} i, \quad E_C = \frac{1}{2} C v_o^2. \quad (51)$$

Here $L_1, C > 0$. The determinant $L_1 L_2 (1 - k^2) > 0$ makes the magnetic store positive definite, including the mutual term once. The exact identity

$$E_m = \frac{1}{2} k L_1 (i_1 + s \rho i_2)^2 + \frac{1}{2} (1 - k) L_1 i_1^2 + \frac{1}{2} (1 - k) L_2 i_2^2 \quad (52)$$

separates the shared magnetizing mode and leakage terms. Taking k to one removes leakage but retains finite magnetizing inductance. At exactly $k = 1$ the differential equations need a constrained formulation; replacing them with (41) is a different model, not that limit alone.

Let $\eta = 1$ for the tapped connection and $\eta = 0$ for the isolated one, using each circuit's local return coordinates. The source voltage u acts through $R_s > 0$, and the effective core-loss conductance $G_c \geq 0$ is across winding 1. Put $G = G_L + G_P$ for load and resistive probe. Then

$$v_a = \frac{u - R_s(i_1 - \eta i_2)}{1 + R_s G_c}, \quad I_s = i_1 - \eta i_2 + G_c v_a, \quad (53)$$

$$\mathbf{L} \dot{i} = \begin{pmatrix} v_a \\ v_o - \eta v_a \end{pmatrix} - \text{diag}(R_1, R_2) i, \quad C \dot{v}_o = -i_2 - G v_o. \quad (54)$$

These circuit equations define the state $x = (i_1, i_2, v_o)^T$ before any work is evaluated. The core resistor is a declared loss path, not a hysteresis law. Resistor heat is exported immediately; no thermal storage is included.

The whole boundary contains the windings, capacitor, source resistor, and core-loss resistor. Its seven powers and their effort–flow meanings are:

Table 8. Complete finite-circuit boundary. The source, receiver, probe, and thermal reservoirs are external. The thermal equalities define immediate export to fixed-temperature reservoirs.

Port	Effort times flow, positive into the boundary
Ideal source	uI_s
Source-resistor heat export	$-(R_s I_s)I_s = -T_s \dot{S}_s$
Winding 1 copper heat export	$-(R_1 i_1)i_1 = -T_1 \dot{S}_1$
Winding 2 copper heat export	$-(R_2 i_2)i_2 = -T_2 \dot{S}_2$
Effective core-resistor heat export	$-v_a(G_c v_a) = -T_c \dot{S}_c$
Receiver	$-v_o(G_L v_o)$
Probe	$-v_o(G_P v_o)$

Multiply the magnetic equation by i^T , the capacitor equation by v_o , and use $u = v_a + R_s I_s$. This independently derives

$$\frac{d}{dt}(E_m + E_C) = uI_s - R_s I_s^2 - R_1 i_1^2 - R_2 i_2^2 - G_c v_a^2 - G_L v_o^2 - G_P v_o^2. \quad (55)$$

The winding-only subboundary instead uses E_m , all winding terminal contributions, and the two copper losses. The shunt core resistor is outside it. Omitting a shifted return in that subboundary gives exactly its missing $\int g I_c dt$ term, with either sign possible; the complete sum obeys (47) at all times.

At $k = 0$, an isolated zero-state secondary satisfies a homogeneous system, so $i_2 = v_o = 0$ exactly. A tapped circuit retains the conductive path through winding 2. Its initial forcing $v_o - v_a$ is generally nonzero, so it can respond without magnetic coupling. This is a topological difference, not a change of voltage label.

6.3 Event intervals and exact finite-state integrals

Use the illustrative baseline $L_1 = L_2 = 1/25$ H, $k = 19/20$, $s = +1$, $C = 1/500$ F, $R_s = 1/5$ Ω , $R_1 = R_2 = 1/10$ Ω , $G_c = 1/100$ S, $G_L = 1/10$ S, and $G_P = 0$. All initial states are zero unless explicitly changed. With t in seconds, the drive is $u = 2 \cos(14\pi t + 3/10)$ V, multiplied by $[1 - \cos(10\pi t)]/2$ during preparation.

Table 9. Four separate integration intervals. Winding currents and capacitor voltage are continuous; powers may jump at the load and source events.

Interval, s	Physical path	Endpoint and event condition
$[0, 1/10]$	Smooth preparation	Starts at $x = 0$; envelope reaches one
$[1/10, 1/5]$	Operation	Same circuit and full drive
$[1/5, 3/10]$	Changed load	G_L increases to $2/5$ S
$[3/10, 1/2]$	Finite relaxation	$u = 0$; source resistance retains a path

Setting the source voltage to zero does not open its conductor. Removing only the load leaves the capacitor. For these finite-path events, integrating the state equations across a vanishing interval gives $x^+ = x^-$ and hence no store jump or model impulse. Source and load powers use their own event-side values. A real switch-control supply is absent from this ideal selector model; its unknown work is not claimed zero on apparatus.

A closed form retains every finite work without a numerical trajectory. On one constant-topology interval write (54) as $\dot{x} = Ax + bu(t)$. Represent the sinusoidal forcing by real oscillator coordinates z satisfying $\dot{z} = Dz$, $u = cz$. A cosine–sine pair at frequency ω has $D_\omega = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$. The preparation waveform is exactly $\cos(14\pi t + 3/10) - \frac{1}{2}\cos(24\pi t + 3/10) - \frac{1}{2}\cos(4\pi t + 3/10)$ in volts, so three such pairs suffice. The full-drive interval needs one pair; constant drive uses a zero-frequency coordinate. With $y = (x, z)$,

$$\dot{y} = Ay, \quad A = \begin{pmatrix} A & bc \\ 0 & D \end{pmatrix}, \quad y(t_0 + T) = e^{TA}y_0. \quad (56)$$

Every effort and flow is a linear functional of y . If they are $e^T y$ and $f^T y$, define $Q = (ef^T + fe^T)/2$, so its signed power is $y^T Q y$. The minus signs of exports belong in e or f .

Theorem 1 (Separate closed-form port works). *For a constant matrix A , define $B = A \otimes I + I \otimes A$ and the entire matrix function $\varphi_1(Z) = \sum_{m=0}^{\infty} Z^m / (m+1)!$. Column vectorization is denoted by vec . Each signed port integral is exactly*

$$\mathcal{W}(Q; A, y_0, T) = \text{vec}(Q)^T T \varphi_1(TB) \text{vec}(y_0 y_0^T). \quad (57)$$

It is valid even when B is singular.

Proof. $Y = yy^T$ obeys $\dot{Y} = AY + YA^T$, whence $\text{vec } Y(t_0 + t) = e^{tB} \text{vec } Y(t_0)$. Integrating its convergent exponential series gives $T \varphi_1(TB)$. Contracting with Q integrates that port's effort–flow product. No endpoint store or balance residual enters this calculation. $\square$

The infinite series defines a closed matrix function, without truncation or small-parameter expansion. An equivalent exact definition is the upper-right block of $\exp\left[T \begin{pmatrix} B & I \\ 0 & 0 \end{pmatrix}\right]$. Linear voltage integrals are likewise $\ell^T T \varphi_1(TA) y_0$. At a physical event, propagate the physical x continuously, initialize the forcing coordinates at the event's actual phase, and apply (57) on the next interval.

For clarity, let $\mathcal{W}_j(\eta, p)$ denote the sum of the four separate closed forms (57) for port j , with probe conductance p . Let $X_{\eta,p}(t)$ be the physical part of the successive exponentials and $\mathcal{E}(X) = i^T Li/2 + Cv_o^2/2$. The complete three-circuit comparison, with every transfer explicitly defined, is:

Table 10. Exact finite expressions, in joules, for $[0, 1/2]$ s. The matrices are fixed by (54), (56), the seven effort–flow pairs, and the interval table. No work is assigned as the remainder of another quantity.

Work or endpoint	Tapped	Isolated	Tapped, 10 Ω probe
Source	$\mathcal{W}_s(1, 0)$	$\mathcal{W}_s(0, 0)$	$\mathcal{W}_s(1, 1/10)$
Source heat	$\mathcal{W}_{R_s}(1, 0)$	$\mathcal{W}_{R_s}(0, 0)$	$\mathcal{W}_{R_s}(1, 1/10)$
Copper 1	$\mathcal{W}_{R_1}(1, 0)$	$\mathcal{W}_{R_1}(0, 0)$	$\mathcal{W}_{R_1}(1, 1/10)$
Copper 2	$\mathcal{W}_{R_2}(1, 0)$	$\mathcal{W}_{R_2}(0, 0)$	$\mathcal{W}_{R_2}(1, 1/10)$
Core heat	$\mathcal{W}_c(1, 0)$	$\mathcal{W}_c(0, 0)$	$\mathcal{W}_c(1, 1/10)$
Load	$\mathcal{W}_L(1, 0)$	$\mathcal{W}_L(0, 0)$	$\mathcal{W}_L(1, 1/10)$
Probe	0	0	$\mathcal{W}_P(1, 1/10)$
Initial store	0	0	0
Final store	$\mathcal{E}(X_{1,0}(1/2))$	$\mathcal{E}(X_{0,0}(1/2))$	$\mathcal{E}(X_{1,1/10}(1/2))$
Exact r_E	0	0	0

This table is not an efficiency ranking: equal source waveform and winding parameters do not match

terminal impedance or output voltage. A resistive probe has its own negative work and changes A , hence all other works and endpoints. It can raise source input or reduce original-load delivery under particular conditions, but neither ordering is asserted as a theorem for all finite transients. Even $G_P = 0$ retains the output capacitor and therefore is not a probe on an otherwise unchanged bare winding.

The finite waveform, store, and signed-work functions are respectively the linear projection of (56), $\mathcal{E}(X(t))$, and (57) with its variable upper time. These exact expressions replace any inference from a plotted numerical trajectory: stores may rise, fall, and remain nonzero after relaxation. In fact a source-free regular interval has $x(t_1) = e^{A(t_1-t_0)}x(t_0)$; the exponential is invertible. A nonzero state therefore cannot reach the exact zero state at finite time through that passive linear relaxation alone.

Whether the different receiving circuits can be prepared and reset with all controller works accounted for is a separate accessible question [OP-TRF-05]. For an explicit capacitor-only reset, a fixed resistor R across an initially charged C gives $v(t) = V_0 e^{-t/(RC)}$. On $[0, T]$ its own heat-port integral is

$$W_R = - \int_0^T \frac{V_0^2}{R} e^{-2t/(RC)} dt = - \frac{CV_0^2}{2} (1 - e^{-2T/(RC)}), \quad E(T) = \frac{CV_0^2}{2} e^{-2T/(RC)}. \quad (58)$$

A voltmeter and resistor-current measurement at ordinary voltage levels resolve the remaining charge and the finite reset work. This distinguishes a finite residual store from a claimed exact reset. An active reset may instead reach zero in finite time, but its supply and switching ports must be included. The elementary reset does not specify how nonzero winding currents were prepared; their separate source and loss paths remain necessary.

6.4 Physical return loading in the finite circuit

For the tapped circuit keep the capacitor across b – c and set $v_b = V_b - V_c$. Let $h = 1$ select a receiver return at a and $h = 0$ at c , with $v_L = v_b - hv_a$ and $G = G_L + G_P$. Current conservation and the source resistor now give

$$I_s = i_1 - i_2 + G_c v_a - hGv_L, \quad (59)$$

$$v_a = \frac{u - R_s(i_1 - i_2) + R_s hGv_b}{1 + R_s(G_c + hG)}, \quad (60)$$

$$Li = \begin{pmatrix} v_a \\ v_b - v_a \end{pmatrix} - \text{diag}(R_1, R_2)i, \quad C\dot{v}_b = -i_2 - Gv_L. \quad (61)$$

The expression for v_a uses $h^2 = h$; h denotes a discrete physical connection, not a continuously interpolated fractional return. Omitting $-hGv_L$ changes the source circuit. The seven powers retain their definitions except for load and probe, which are $-G_L v_L^2$ and $-G_P v_L^2$. With $j_L = G_L v_L$, their load-terminal decomposition is

$$P_{L,b} = -v_b j_L, \quad P_{L,\text{ret}} = hv_a j_L, \quad P_{L,b} + P_{L,\text{ret}} = -G_L v_L^2. \quad (62)$$

Integrating each of these before combining verifies the decomposition. The same multiplication proof as for (55) gives zero exact energy residual with stores $E_m + Cv_b^2/2$ for either h .

The nominal paired example has $n = -4$, $k = 19/20$, and $G_L = 1/10$ S. Its operation interval $[1/10, 1/5]$ s has the exact observations, with $T = 1/10$ s,

$$\Psi_L^{(h)} = \ell_h^T T \varphi_1(TA_h)y_h, \quad W_j^{(h)} = \mathcal{W}(Q_{jh}; A_h, y_h, T), \quad (63)$$

$$E_0^{(h)} = \mathcal{E}(x_h), \quad E_1^{(h)} = \mathcal{E}(X_h(1/5)). \quad (64)$$

Here $x_h = X_h(1/10)$ is independently prepared by its own first-interval matrix, y_h augments that state with the forcing phase, and $\ell_h^T y = v_L$. The two initial operation stores need not agree even though both preparations start from zero. Subtracting only already integrated works gives the exact paired balance $\Delta E^{(1)} - \Delta E^{(0)} = \sum_j (W_j^{(1)} - W_j^{(0)})$.

There is no general ordering of the load deliveries. Two exact initial-state arguments already show opposite signs. At $x = (0, 0, 0)$ with $u \neq 0$, return a has $v_L = -u/[1 + R_s(G_c + G)]$, whereas return c has $v_L = 0$. At $x = (0, 0, V_0)$ with $u = 0$, return a has $v_L = V_0(1 + R_s G_c)/[1 + R_s(G_c + G)]$, smaller in magnitude than the return- c value V_0 when $R_s G > 0$. Continuity preserves these strict inequalities on some nonzero following interval when $G_L > 0$. At $G_L = 0$ both receiver works are zero. The prepared-state example specifies observation endpoints, not the missing preparation work.

On every one of these finite trajectories the additive readout relation is exact. The ideal winding ratio generally is not. Define $n = s\rho$ and $D_\Psi = \Psi_{ba} - n\Psi_{ac}$. Integrating both constitutive winding equations independently yields

$$D_\Psi = \Delta(\lambda_2 - n\lambda_1) + \int_{t_0}^{t_1} (R_2 i_2 - nR_1 i_1) dt, \quad \lambda = Li. \quad (65)$$

The linkage endpoints are obtained from the state, while each copper integral has its own linear primitive. A nonzero D_Ψ is a finite-model voltage observation, not an energy residual. Its sign depends on state and interval.

6.5 Parameter distinctions and constrained limits

The regular derivations cover both connections and both polarities. In particular, they include

$$\rho \in \{1/2, 1, 2\}, \quad k \in \{0, 1/2, 19/20, 99/100\}$$

without any interpolation assumption. The additional individually specified controls are important because they alter different parts of the equations:

Table 11. Individual regular controls, with other baseline parameters retained. Their joint physical realization is not established by the algebraic balance alone.

Parameter or condition	Alternatives to the baseline
Turns magnitude	1/4, 4
Initial load conductance, S	0, 1/1000, 1
Probe conductance, S	1/1000, 1/10
Drive frequency, Hz	0, 2, 20
Capacitance, F	1/2000, 1/100
Both copper resistances, Ω	0, 1/2
Core conductance, S	0, 1/10
Load multiplier at 1/5 s	0 (removal), 1 (no change)
Drive phase	$-\pi/2, 0, \pi/2$
Imposed (i_1, i_2, v_o) , in A, A, V	(1/5, 0, 0); (0, -1/5, 0); (0, 0, 1)
Null	Zero source and zero initial state

The paired-return family additionally uses opposed $\rho \in \{3, 4, 6\}$, the same four k values, $G_L \in \{0, 1/10, 1\}$ S, and both returns. Its separate $n = -4$ controls are zero and 20 Hz drive, initial $(1/5, 0, 0)$ or $(0, 0, 1)$, $G_P = 1/10$ S, reversed polarity, zero drive, and equal opposed sections $n = -1$. All four physical intervals remain separate; neither fixed-return experiment itself commutates the return.

Exactly $k = 1$, $C = 0$, $R_s = 0$, a shorted output, and zero turns ratio are not obtained by substituting into a formula that divides by the vanishing quantity. They need their own constrained equations and compatible initial states; a prepared capacitor short may require an impulse or a dissipative finite path. Saturation, remanence, hysteresis, frequency-dependent core loss, distributed capacitance, chassis returns, and thermal evolution are absent from (54). Their model residual has unknown sign and magnitude. No equality of ideal balances estimates that discrepancy.

7 Compound circulation and a transformer network

7.1 Two meshes and a common stepped planet

Let coaxial members A, B mesh with two rigidly connected planet sections P_A, P_B on carrier C . Set $s_X = -1$ for an external mesh and $s_X = +1$ for an internal mesh, and define

$$a = \frac{m_0}{2} (Z_A - s_A Z_{pA}) = \frac{m_0}{2} (Z_B - s_B Z_{pB}) > 0, \quad h_X = s_X \frac{Z_{pX}}{Z_X}, \quad R = \frac{h_A}{h_B}. \quad (66)$$

Rolling gives $\omega_X - \omega_C = h_X(\omega_P - \omega_C)$ and hence $\omega_A - \omega_C = R(\omega_B - \omega_C)$. With carrier rate $\omega_C = w$ and one held member, the other rates are

	ω_A	ω_B	$\omega_P = w + \frac{\omega_A - w}{h_A}.$
A held	0	$w(1 - 1/R)$	
B held	$w(1 - R)$	0	

(67)

For q equally loaded massless planets, choose $\tau_A = T$. Their axial free-body equation is $s_A r_{pA} F_A + s_B r_{pB} F_B = 0$, with $q r_A F_A = T$. The pin reaction is $R_t = -(F_A + F_B)$, giving

$$(\tau_A, \tau_B, \tau_C) = T(1, -R, R - 1). \quad (68)$$

The carrier torque follows from $q a R_t$ using (66). These constraints, not an energy premise, imply $\sum \tau = 0$ and $\sum \tau \omega = 0$.

The nine named sides are three shaft powers $p_X^f = \tau_X(\omega_X - \Omega)$, two opposite sides of each mesh, and two opposite pin sides. When all planets are aggregated, the mesh sides are $\pm p_A^f, \pm p_B^f$ and the pin sides are $\pm p_C^f$. They are component diagnostics, not nine independent external supplies. Fix the physical ground throughput denominator $D_0 = \max_{X=A,B,C} |\tau_X \omega_X|$ and define

$$\mathcal{E}_f = \frac{\max_{\text{nine sides}} |p^f|}{D_0} = \frac{\max_{X=A,B,C} |\tau_X(\omega_X - \Omega)|}{D_0}. \quad (69)$$

This definition is meaningful only for $D_0 > 0$ and gives $\mathcal{E}_0 = 1$ exactly. It measures the largest absolute indicated port power, not a new net transfer and not an invariant amount of circulating energy. In particular,

$$\mathcal{E}_C = \frac{1}{|R - 1|} \quad (A \text{ held}), \quad \mathcal{E}_C = \frac{|R|}{|R - 1|} \quad (B \text{ held}). \quad (70)$$

Table 12. Five pitch-compatible compound configurations. Multiple-planet tooth phasing, clearance, and equal load sharing remain additional physical assumptions.

Configuration	$(Z_A, Z_{pA}, Z_{pB}, Z_B)$	(s_A, s_B)	R
I: simple planetary	(24, 18, 18, 60)	$(-1, +1)$	$-5/2$
II: stepped planet and ring	(20, 22, 20, 62)	$(-1, +1)$	$-341/100$
III: two external members	(30, 20, 19, 31)	$(-1, -1)$	$62/57$
IV: two rings	(62, 20, 21, 63)	$(+1, +1)$	$30/31$
V: two rings, close ratios	(100, 58, 59, 101)	$(+1, +1)$	$2929/2950$

For each choose $m_0 = 1/500$ m, $q = 3$, $w = 2\pi$ rad/s, $T = 47/4$ N m, and interval $[0, 3/2]$ s. All four frame ratios follow directly from (67)–(69):

Table 13. All held-member and frame choices for the five configurations. The smallest value is 100/441 and the largest is 2950/21. Neither is a continuous bound for other geometries or frames.

Configuration	Held	Ground	Carrier	Member A	Member B
I	A	1	$2/7$	1	$2/5$
I	B	1	$5/7$	$5/2$	1
II	A	1	100/441	1	100/341
II	B	1	341/441	341/100	1
III	A	1	$57/5$	1	$57/62$
III	B	1	$62/5$	$62/57$	1
IV	A	1	31	1	$31/30$
IV	B	1	30	$30/31$	1
V	A	1	2950/21	1	2950/2929
V	B	1	2929/21	2929/2950	1

Each signed work is separately $W_X^f = (3/2)\tau_X(\omega_X - \Omega)$; internal works carry the corresponding opposite signs. This ideal algebraic boundary has no constitutive storage coordinates, so both endpoint store sums are empty. The external works sum to zero and $r_E = 0$ without an impulse. Adding finite inertias would require independent kinetic endpoints and, for acceleration, different torques. Configuration I reduces exactly to the earlier simple train.

At $R = 1$, the constraints impose $\omega_A = \omega_B$, not a stopped carrier. With either member held, both are zero, $\tau_C = 0$, and $D_0 = 0$ even if $w \neq 0$. A nonzero relative circulation effort can still be imposed in the ideal network. Its ratios are then nonzero divided by zero, or zero divided by zero; both are undefined rather than finite observations. For any positive rational $R = p/q$, the two-external assembly $(Z_A, Z_{pA}, Z_{pB}, Z_B) = (2q, 2p, p+q, p+q)$ realizes that ratio with positive integer teeth. Thus approaches from either side of one are algebraically possible, subject again to physical tooth and clearance constraints.

7.2 Complete ports and mapped terminal products

Use two ideal transformers with primary branches $A-C$ and $B-C$ and secondary branches both $P-C$. For each transformer impose

$$V_X - V_C = h_X(V_P - V_C), \quad j_X = -h_X i_X, \quad j_A + j_B = 0, \quad i_C = -(i_A + i_B). \quad (71)$$

The parallel-secondary current law gives $i_B = -Ri_A$; the common return then gives $i_C = (R - 1)i_A$. With the power-preserving scale $\alpha = 1 \text{ V}/(\text{rad/s})$, these reproduce (67)–(68). The conversion from revolutions per second includes 2π in every power and work.

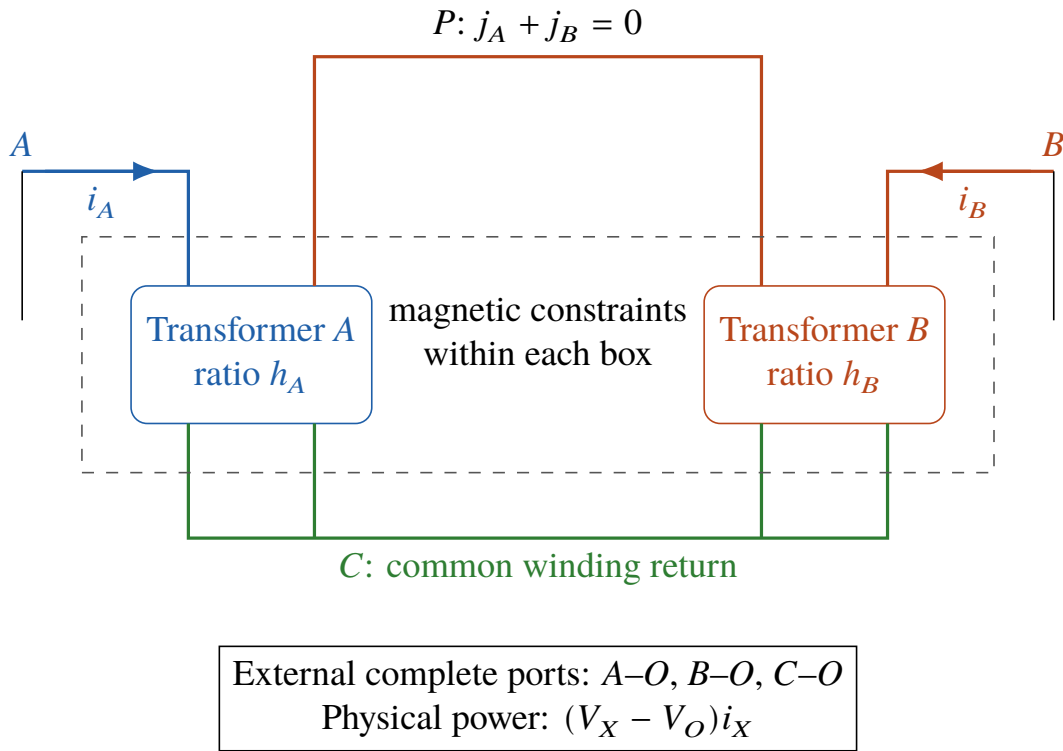


Figure 5. Schematic two-transformer compound correspondence. Both secondary windings share $P-C$; the complete external ports connect A, B, C to the physical return O . The four winding ports are internal to the complete network. A common shift of all node potentials, including O , preserves every complete port voltage.

On the whole electrical boundary the three external ports are $X-O$, with power $(V_X - V_O)i_X$. On individual transformer boundaries the four complete winding powers are $(V_X - V_C)i_X$ and $(V_P - V_C)j_X$; each pair sums to zero. The nine mechanical sides correspond instead to signed copies of the terminal products $(V_X - \alpha\Omega)i_X$. They therefore match exactly in every frame in the table, but they are not all complete electrical ports. For a coordinate shift, an individual external-port return contributes $-(V_O + g)i_X$, even when the sum of all return currents is zero.

The maximum winding power divided by D_0 equals $\mathcal{E}_C$. The ratio including all three external and four winding ports is

$$\mathcal{E}_{\text{phys}} = \max(1, \mathcal{E}_C), \quad (72)$$

which is invariant under a common coordinate shift and generally differs from $\mathcal{C}_f$. For the table, $\mathcal{C}_{\text{phys}} - \mathcal{C}_f$ ranges from $-241/100$ to $2929/21$, and differs from zero for 26 of the 40 frame choices. These are differences between defined observables, not failed matches of the corresponding terminal and mechanical products. A fixed physical-port set cannot realize all frame ratios through changing only the voltage zero. Constant ideal winding voltage here does not establish a sustainable finite-core DC transformer.

The same correspondence retains zero velocity with nonzero torque, zero current, sign reversals, and exact current zeros. For example replace T by $T_*(t - 3/4) \text{ s}^{-1}$ on $[0, 3/2] \text{ s}$ with constant rates. Each signed work vanishes over the full interval, while its two subinterval integrals have opposite values proportional to $\int_0^{3/4} (t - 3/4) dt = -9/32 \text{ s}^2$ and $+9/32 \text{ s}^2$. Instantaneous absolute power and absolute integrated work remain nonzero. Zeros and maximizing-port ties follow from the individual polynomial products; undefined zero-throughput ratios must be retained. Arbitrary constant and linear common offsets, including the earlier nine choices, change terminal products according to (47) while leaving the complete-port works fixed.

8 Finite correspondence and its physical obstruction

8.1 Magnetic elasticity and capacitive inertia

A finite compound network can have real stored states without being a rigid gear train. Let the oriented branch-incidence matrix B have columns $e_A - e_C, e_P - e_C, e_B - e_C, e_P - e_C$, with the held node eliminated when necessary. Each coupled pair has

$$\mathbf{L}_X = L_0 \begin{pmatrix} h_X^2 & kh_X \\ kh_X & 1 \end{pmatrix}, \quad L_0 > 0, \quad 0 \leq k < 1. \quad (73)$$

Let C be the diagonal grounded node-capacitance matrix and R the positive branch-resistance matrix. The electrical model is

$$C\dot{v} = I_{\text{ext}} - Bi, \quad Li = B^T v - Ri. \quad (74)$$

Source, receiver, and probe conductances are included explicitly in I_{ext} , each with its own physical return. Independently specify a rotational Maxwell model with inertias $J = \alpha^2 C$, elastic deformations z , stiffness $K = \alpha^2 L^{-1}$, and mobility $M = R/\alpha^2$. Its equations are

$$J\dot{\omega} = \tau_{\text{ext}} - Bf, \quad f = Kz, \quad \dot{z} = B^T \omega - Mf. \quad (75)$$

For a single pair set $u_z = z_1/h_X$, $v_z = z_2$, and $q_{\pm} = (u_z \pm v_z)/\sqrt{2}$. The elastic energy is

$$U_X = \frac{\alpha^2 q_+^2}{2L_0(1+k)} + \frac{\alpha^2 q_-^2}{2L_0(1-k)}, \quad f = \frac{\partial U}{\partial z}. \quad (76)$$

Differentiation verifies the specified stiffness. The exact map

$$v = \alpha \omega, \quad i = f/\alpha, \quad z = Li/\alpha, \quad I_{\text{ext}} = \tau_{\text{ext}}/\alpha \quad (77)$$

substitutes (74) into (75). Compatible initial states and uniqueness of the regular linear equations establish correspondence throughout each interval. The magnetic store equals U , and capacitor energy equals rotational kinetic energy. Magnetic energy maps to elasticity, while inertia maps to capacitance; interchanging those assignments would invalidate the comparison. Residuals in amperes, volts, radians, and

radians per second cannot be merged into a dimensionful physical norm without scales.

One illustrative coefficient choice is $L_0 = 3/25$ H, $R_X = (2/25) \text{diag}(h_X^2, 1) \Omega$, $R_s = 3/4 \Omega$, with member inertias $3/100, 7/200 \text{ kg m}^2$, carrier inertia $1/50 \text{ kg m}^2$, three planets each of mass $3/25 \text{ kg}$ and spin inertia $1/1250 \text{ kg m}^2$. Include their orbital inertia $3m_p a^2$ separately, using (66). Grounded receiver and driver capacitors $1/100$ and $1/200$ F parallel the free-member and carrier capacitors; a reset conductance is $1/2$ S and source amplitude 2π V at $\alpha = 1$ in the stated units. The branch stores may be counted as four shared/leakage modal stores, and the free-member, carrier, planet, receiver, and driver stores as five more. The held-member capacitor has identically zero voltage. All nine nontrivial terms are evaluated constitutively at both endpoints.

Finite elasticity introduces effort paths absent in the rigid reduction. For example $m_X = j_X + h_X f_X$, using mechanical secondary and primary efforts, is generally nonzero and supplies a couple between planet and carrier. The pin effort includes orbital acceleration,

$$T_{\text{pin}} = J_{\text{orb}} \dot{\omega}_C + (h_A - 1)f_A + (h_B - 1)f_B. \quad (78)$$

Here primary and secondary efforts are the components of f in (75), and $J_{\text{orb}} = 3m_p a^2$. Consequently opposite diagnostic mesh-side products need not cancel across an elastic element. Matching the original nine diagnostics is not proof that those alone exhaust a finite boundary.

8.2 Physical ports, event schedules, and ratios

The complete finite boundary contains all the above stores and source and branch resistors, with external source, receivers, probe, reset, and heat reservoirs. There are ten physical powers: source uI_s ; source heat $-(R_s I_s)I_s$; four winding heat exports $-(R_j i_j)i_j$; ground-return receiver $-(V_F - V_O)G_O(V_F - V_O)$; carrier-return receiver $-(V_F - V_C)G_C(V_F - V_C)$; probe $-(V_F - V_O)G_P(V_F - V_O)$; and reset $-(V_F - V_O)G_R(V_F - V_O)$. F names the free member. Each resistance also has its equivalent $-T\dot{S}$ export. The mechanical forms follow (45), with mobility dissipation $-f^T M f$. Node and winding subboundary powers remain separate diagnostics. Multiplying the two network equations by v^T and i^T cancels $v^T B i$, deriving the energy balance and the corresponding mechanical one.

Preparation $[0, 1/10]$, operation $[1/10, 1/5]$, post-connection change $[1/5, 3/10]$, relaxation $[3/10, 1/2]$, and reset $[1/2, 1]$ s are distinct stages. An explicitly prescribed reversed-drive preparation is also a distinct path. A receiver can remain at $F-O$ or change to $F-C$ while every capacitor return stays at O . Ideal selectors have zero voltage when closed and zero current when open; a stipulated zero-current controller then has zero work within that selector model. This says nothing about a real actuator supply. With bounded resistive changes, all capacitor voltages, winding currents, and elastic deformations are continuous; no inductor is opened, capacitor reattached, or energized rotor clamped.

For constant connections and a forcing represented as in (56), each of the ten works is the separate closed form (57); endpoints use each model's own states and stores. A finite transition of width $1/1000$, $1/2000$, or $1/4000$ s requires a declared conductance history on that transition. The instantaneous identity still follows from the differential equations, but finite numerical results for those histories are not exact closed-form results of this article. They are conditional extensions until their individual path integrals are specified. No limiting switch work is inferred from the instantaneous model.

Linearity gives an exact amplitude control: reversing the source and the compatible prepared state reverses all physical states and leaves their quadratic physical powers and works unchanged. Zero drive with zero state is the exact null. Common-offset cross terms gI_j do reverse with current and must be recomputed; they are not copied from positive drive. Neither amplitude homogeneity nor a regular finite

model at $R = 1$ proves the singular rigid constraint or an energized locking law.

In fixed ground coordinates, define the instantaneous diagnostic denominator on the actual trajectory by

$$D(t) = \max_{X=A,B,C} |V_X I_{\text{ext},X}|. \quad (79)$$

A receiver commutation does not authorize replacing this with the old circuit's counterfactual trajectory. The mapped nine-side ratio uses its own numerator; a complete-physical-port ratio uses another numerator. An interval ratio instead first integrates every signed work and then takes the prescribed maxima and quotient. It is neither the instantaneous ratio nor an integral of absolute power. Zeros, nonzero-over-zero, and zero-over-zero remain undefined; a small whole-boundary residual cannot certify a small or uncertain denominator.

An initially unenergized right-hand operation endpoint can have zero node throughput while the external source and its resistor already carry finite, opposite powers. Thus the mapped diagnostic can be zero-over-zero while the complete physical diagnostic is nonzero-over-zero. There is no finite ratio to report at that event. In a finite prepared example, ratios at 3/20, 1/4, and 3/4 s are obtained by evaluating its exact state and individual powers there, not by reusing (70). Inertia, compliance, resistance, loading, and excitation all change them. No ordering or extremum over those stages follows from the rigid table.

8.3 An accelerating reference is not a voltage coordinate shift

For the finite rotational analogue, the relative kinetic-plus-elastic store is $K^f + U = K^0 - \Omega H + J\Omega^2/2 + U$, with $J = \sum J_j$ and $H = \sum J_j \omega_j$. Its additional Euler power is $-\dot{\Omega}(H - J\Omega)$, without the centrifugal potential convention used for E^f in (18). Either convention is valid when its terms are retained consistently. Relative deformations and mobility losses are unchanged under common rigid rotation.

A common electrical coordinate shift, however, also shifts V_O and leaves all actual capacitor voltage differences and physical energies unchanged. The difference between the mechanical relative store and that unchanged physical electrical store is exactly

$$\Delta E_{\text{ref}} = -\Omega H + \frac{1}{2} J \Omega^2. \quad (80)$$

An effective electrical expression built from shifted node values relative to an unshifted fictitious zero can imitate it, but then represents a virtual store rather than the actual grounded capacitors. Realizing moving capacitor returns requires new driven connections and their supplies. Their work and hardware discrepancy are unknown. Equation (80) locates a difference of store definitions and does not justify adding an invented reference-energy transfer to the physical circuit.

Consequently the regular compliant correspondence is exact, while full rigid loaded duality, independently driven reference motion, actual reference commutation, and linkage-phasing realization remain open physical questions. Finite $k \rightarrow 1$ with fixed inductances, inertias, and resistances does not recover the rigid quasistatic boundary automatically. Exact shorting, zero-capacitance constraints, zero source resistance, prepared locking, and finite full-interval zero-net-work controls need separate formulations. The ideal counterpart remains useful within its own stated scope.

9 Events, uncertainty, and limits of the conclusions

9.1 Discontinuities and impulses

A mesh separation or re-engagement with finite force can have continuous positions and velocities and hence zero store jump, while its force-velocity product jumps. The exact work is the sum of integrals

on both event sides; no additional impulse is inserted. A compliant contact example is an overlap δ with spring force $k_m \max(\delta, 0)$ and stored energy $k_m \max(\delta, 0)^2/2$. Adding a dashpot requires its own non-negative relative-motion heat export. Switching a finite contact or damping force does not itself establish a jump in momentum.

Rigid impact with restitution is different: momenta and kinetic energy may jump. Its signed impulse work requires a declared impact law or finite regularization. Backlash during an accelerating carrier, unequal planet load sharing, nonzero pin couples, and radial centre motion cannot be certified by the simpler event or steady-state formulas. Likewise, physically opening an energized winding can produce clamp, arc, and displacement-current paths; changing a tap or moving a capacitor can involve charge sharing. Every switch, clamp, controller, and preparation path must then be explicit. A finite resistive receiver selector does not settle these reactive events.

9.2 Exact residuals and unresolved accuracy questions

All polynomial, trigonometric cycle, and matrix-function integrals reported here are exact in their declared models. The numerical integration residual is **exactly zero**, because no numerical integration is used. The instantaneous equation residual is zero by the independent constitutive substitution shown for each model, and the exact energy residual is zero only for the complete boundaries and effective-frame conventions specified. The physical model residual remains unquantified: it comprises omitted mesh, bearing, windage, churning, joint, magnetic, dielectric, switch, controller, and thermal physics. No apparatus measurements are reported.

It is useful to retain the mathematical limitations of approximate audits without turning their outcomes into physical theorems. Three distinct failure mechanisms illustrate why an apparently small balance is insufficient. First, a quadrature error estimate can vanish for a constant integrand while roundoff in a nonzero work sum remains. Second, an accumulation estimate proportional to the computed power can miss cancellation error when forming $\omega - \Omega$ near an exact zero. If the represented rates have errors $\delta\omega, \delta\Omega$, the port-power error at the exact held condition is exactly $\tau(\delta\omega - \delta\Omega)$ for exact τ ; its work can have either sign. Reducing an integration step cannot remove that input error. Zeroing the torque removes it for a different reason and is a useful but distinct null control.

For N contributions combined by a balanced addition tree of depth $d = \lceil \log_2 N \rceil$, the standard elementary relative-error model $|\delta| \leq u$, without underflow or overflow, gives the conservative exact bound $[(1+u)^d - 1] \sum_{j=1}^N |a_j|$ for summation error. This follows by bounding each leaf's product of at most d relative factors. It still bounds summation of the given a_j , not the error in constructing them. A first-order du replacement is unnecessary here. Completing a missing error component must be justified independently; changing a criterion only to admit a residual provides no validation.

Third, a boolean statement $|r_E| \leq b$ is unstable when its uncertainty spans the threshold. With certified residual uncertainty u_r , only $|r_E| + u_r \leq b$ establishes acceptance; when $|r_E| - u_r > b$ it establishes failure; the intervening case is unresolved. Comparing a small positive bound with a much larger absolute comparison tolerance can be vacuous even if nothing is known to disagree. These questions concern classification and detection, not necessarily energy transfer. The exact nulls in (30) and (41) settle those ideal equations but do not retrospectively certify any approximate calculation or unknown implementation.

Finite-circuit work uncertainty, finite voltage-integral uncertainty, and finite compound ratio uncertainty remain separate. As explicit illustrative engineering decision limits, one may declare

$$b_W = \frac{1}{5000000} J + \frac{1}{500000} \sum_j |W_j|, \quad (81)$$

$$b_{\Psi} = \frac{1}{50000000} V s + \frac{1}{500000} \sum_l |\Psi_l|. \quad (82)$$

The second sum covers the three readouts and the two separately defined copper-drop voltage integrals. These fixed criteria are not error enclosures. Uncertainty for a fixed-output circuit and for two separately loaded returns are separate questions; checking one cannot certify the other. A joule discrepancy must not be added to a volt-second discrepancy. Different integration estimates may disagree while an equation-based energy residual is small through cancellation. Power-zero locations and maximum-port switches require complete root reasoning for the continuous functions; a collection of sign brackets alone is not such a proof, and an identically zero interval is not a set of isolated crossings. Sensitivity evidence is not a rigorous enclosure or a hardware model-error estimate. An uncertainty interval for a denominator that contains zero leaves its ratio unresolved. None of these unresolved accuracy questions is closed merely by the exact identities of a different idealization.

9.3 Physical scope

The mechanical derivation assumes parallel-axis rigid pitch geometry, compatible assembly, constant centre radius, equal loading, and zero pin couple. Ravigneaux arrangements require further independent mesh equations; helical or nonparallel-axis motion requires additional degrees of freedom. Bodies with flexure, true restitution, combined backlash and Euler forcing, and rate reversals in frictional mechanisms require their own constitutive and event treatments. The kinematic count identities themselves remain valid through reversals, while absolute counts and friction signs need interval splits. Tooth counts are integers; irrational rate ratios are allowed by the linear constraints and are not a new energy law.

The lead-out model uses massless coupling members and a resolved axial torque path only. It omits clearance, compliance, encoder quantization and direction ambiguity, bearing friction, joint-force and bending work, and the energetic cost of the ground-phase device. Its illustrative large articulation is not a buildability claim. A combined train and readout must recompute the forces and trajectory under the actual load. Conditions such as a zero pin flow follow from geometry; force and work magnitudes depend on the selected inertias and torques. Inertial contributions scale linearly with their mass coefficient for a fixed prescribed trajectory, and quasistatic terms scale linearly with applied torque; a loaded trajectory need not retain either scaling when those parameters change.

For the windings, common-coordinate invariance proves nothing about a new physical ground path. Regular linear closure proves nothing about nonlinear core or thermal behavior, singular limits, real switching supplies, missing preparation, or exact reset. Individual parameter controls do not validate all joint physical parameter combinations. A calibrated hardware comparison must retain the source, receiver, probe, all returns, each energy store, and instrument loading over the same event-split interval. Any unexplained signed remainder then remains an open result with its magnitude, sign, conditions, and completed checks; an expected conservation law is not a substitute for that investigation.

10 Conclusion

Signed effort–flow integrals distinguish a change of observation from a change of apparatus. In the gear example, the sun’s $987\pi/8$ J ground work becomes $705\pi/8$ J in carrier coordinates, accompanied by an explicit held-ring contribution and the appropriate frame stores. The locked zero-drive control changes the body count while every physical port work stays zero. The history-dependent split of frame terms follows (29), not a universal equal-partition rule.

A common electrical voltage shift preserves complete port powers and moves only their terminal

decomposition. The exact missing-return residual $21/4$ J and the loaded 16 J versus 9 J example follow from separately integrated products. Finite loading changes both trajectories and endpoint stores; it cannot be inferred from volt-seconds.

The compound maximum-power ratios $100/441$ through $2950/21$ belong to specified observations, while their ground-reference values are one. The ideal terminal map and the regular compliant state map have exact stated scopes. A physical accelerating electrical reference, fully loaded lead-out reaction paths, reactive commutation, and hardware discrepancies remain open; none is supplied by a coordinate identity or by an undeclared transfer.

References

1. Hermann A. Haus and James R. Melcher (1989), *Electromagnetic Fields and Energy*, Prentice Hall, Englewood Cliffs, NJ. Section 9.7, “Magnetic Circuits,” especially “Electrical Terminal Relations and Characteristics.” [Author text hosted by MIT](#).